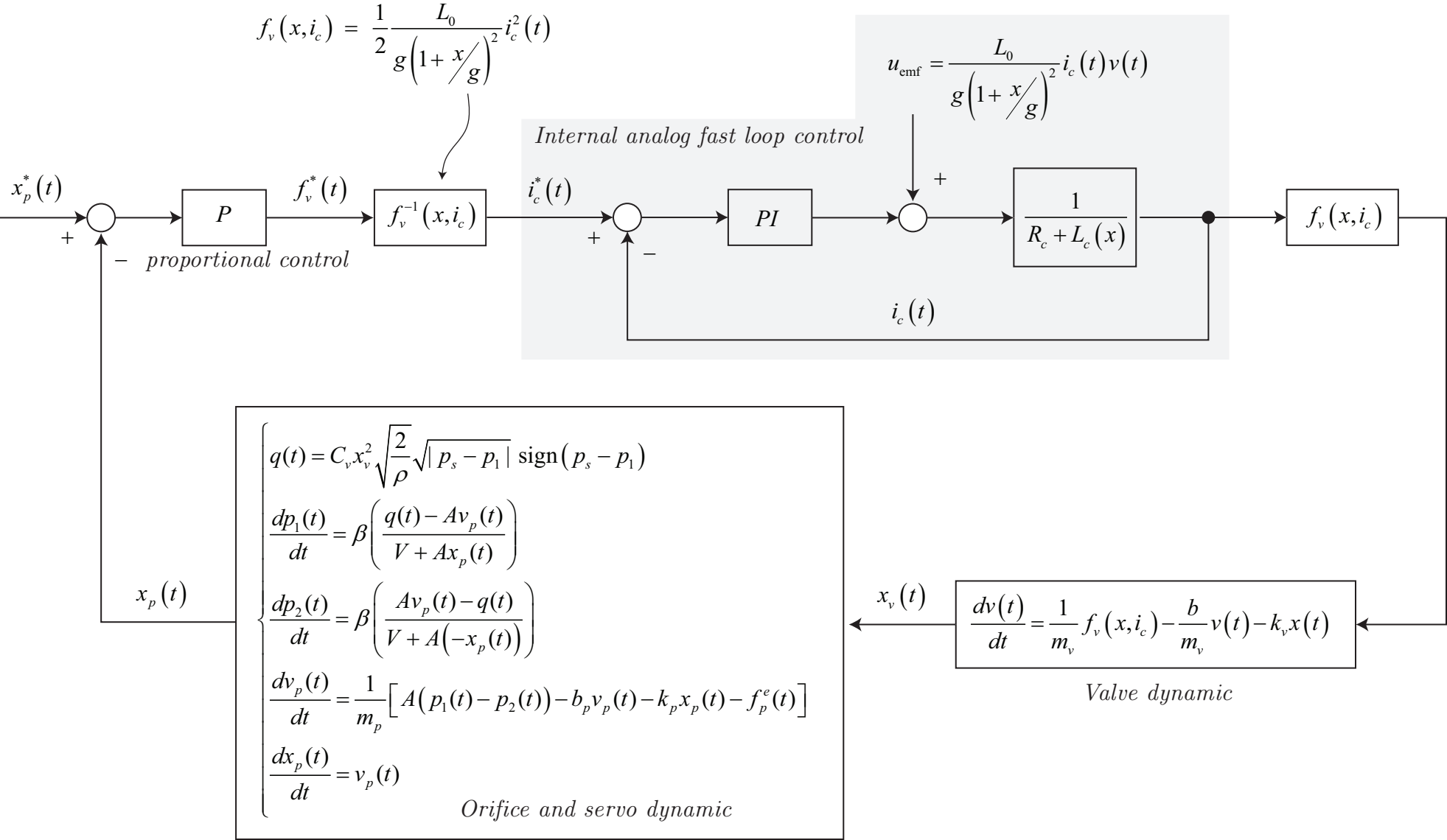
$f_v(x,i_c) = \frac{1}{2}\frac{L_0}{g\left(1+x/g\right)^2} i_c^2(t)$
$u_{\text{emf}} = \frac{L_0}{g\left(1+x/g\right)^2} i_c(t) v(t)$
Internal analog fast loop control
$x_p^*(t)$
+
−
P
proportional control
$f_v^*(t)$
$f_v^{-1}(x,i_c)$
$i_c^*(t)$
+
−
PI
+
$\frac{1}{R_c + L_c(x)}$
$i_c(t)$
$f_v(x,i_c)$
$x_p(t)$
$q(t) = C_v x_v^2 \sqrt{\frac{2}{\rho}} \sqrt{|p_s - p_1|}\ \text{sign}(p_s - p_1)$
$\frac{dp_1(t)}{dt} = \beta\left(\frac{q(t) - Av_p(t)}{V + Ax_p(t)}\right)$
$\frac{dp_2(t)}{dt} = \beta\left(\frac{Av_p(t) - q(t)}{V + A(-x_p(t))}\right)$
$\frac{dv_p(t)}{dt} = \frac{1}{m_p}\left[A(p_1(t) - p_2(t)) - b_p v_p(t) - k_p x_p(t) - f_p^e(t)\right]$
$\frac{dx_p(t)}{dt} = v_p(t)$
Orifice and servo dynamic
$x_v(t)$
$\frac{dv(t)}{dt} = \frac{1}{m_v} f_v(x,i_c) - \frac{b}{m_v} v(t) - k_v x(t)$
Valve dynamic